

Literals

In [5]: `# Numerical Literals`

```
print(5)
```

5

In [10]: `print(type('5.8'))`

```
<class 'str'>
```

In [11]: `a = 5`
`print(a)`

5

In [12]: `"abc"`

Out[12]: 'abc'

In [14]: `type(True)`

Out[14]: bool

In [15]: `5 == 6`

Out[15]: False

In [16]: `5 == 5`

Out[16]: True

In []: ▶

Variables

In [18]: ▶

```
a = 5  
print(a)
```

5

In [19]: ▶

```
b = 6  
b
```

Out[19]: 6

In [20]: ▶

```
c = 6.5  
c
```

Out[20]: 6.5

In [21]: ▶

```
d = 'sourav'  
d
```

Out[21]: 'sourav'

In [22]: ▶

```
e = True  
e
```

Out[22]: True

```
In [26]: f = (2 + 5j)
         f
```

```
Out[26]: (2+5j)
```

```
In [27]: type(f)
```

```
Out[27]: complex
```

```
In [ ]:
```

Identifiers

```
In [28]: ab12_ = 56
         ab12_
```

```
Out[28]: 56
```

```
In [29]: _ab12 = 67
         _ab12
```

```
Out[29]: 67
```

```
In [30]: 12ab_ = 8
         12ab_
```

```
Cell In[30], line 1
```

```
    12ab_ = 8
      ^
```

```
SyntaxError: invalid decimal literal
```

```
In [31]:  a = 5  
         a
```

Out[31]: 5

```
In [32]:  _ = 9  
         _
```

Out[32]: 9

```
In [33]:  7 = 8  
         7
```

Cell In[33], line 1

```
  7 = 8  
    ^
```

SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of '='?

```
In [34]:  7 == 8
```

Out[34]: False

```
In [ ]: 
```

```
In [35]:  a = 99  
         a
```

Out[35]: 99

```
In [39]: A = 99
A
```

```
Out[39]: 99
```

```
In [ ]:
```

```
In [37]: print(a)
```

```
99
```

```
In [40]: a is A
```

```
Out[40]: True
```

```
In [41]: if = 8
if
```

```
Cell In[41], line 1
```

```
if = 8
```

```
^
```

```
SyntaxError: invalid syntax
```

```
In [42]: def = 9
def
```

```
Cell In[42], line 1
```

```
def = 9
```

```
^
```

```
SyntaxError: invalid syntax
```

```
In [43]: ▶ cjsbcjsbcjcmnbcbsbchscjsbchsbcbjhdscbvshdcsnvcjsbcsbcmnbvjd = 9  
cjsbcjsbcjcmnbcbsbchscjsbchsbcbjhdscbvshdcsnvcjsbcsbcmnbvjd
```

Out[43]: 9

```
In [ ]: ▶
```

Keywords

```
In [53]: ▶ raj = 95  
  
if raj > 90:  
    print('Raj will be eleigible to take certificate.')  
else:  
    print('Not Eligible')
```

Raj will be eleigible to take certificate.

```
In [51]: ▶ raj > 90
```

Out[51]: False

```
In [52]: ▶ 67 > 90
```

Out[52]: False

```
In [54]: ▶ a = 1  
a
```

Out[54]: 1

```
In [55]:  a = 2
          a
```

```
Out[55]: 2
```

```
In [63]:  lst = [2,3,4,5,7,8]
          lst
```

```
Out[63]: [2, 3, 4, 5, 7, 8]
```

```
In [59]:  for i in lst:
          {
              print(i)
          }
```

```
2
3
4
5
7
8
```

```
In [66]:  lstt = []
          for i in lst:
              if i%2 == 0:
                  lstt.append(i+1)

          lstt
```

```
Out[66]: [3, 5, 9]
```

```
In [67]:  lst
```

```
Out[67]: [2, 3, 4, 5, 7, 8]
```

In [68]: ▶ lstt

Out[68]: [3, 5, 9]

In []: ▶

In []: ▶

Comments

In [75]: ▶ *# Single Line Comment*
here we basically filtering the even values

```
lstt = []  
for i in lst:  
    if i%2 == 0:  
        lstt.append(i+1)  
  
lstt
```

Out[75]: [3, 5, 9]

In []: ▶


```
In [77]: """
Multi Line Comment

here we are basically filtering the even numbers
and then we are adding one more value

"""

lstt = []
for i in lst:
    if i%2 == 0:
        lstt.append(i+1)

lstt
```

Out[77]: [3, 5, 9]

In []:

Operators

Arithmetic Operators

```
In [78]: a = 8
b = 9

c = a + b
print(c)
```

17

```
In [79]: ▶ a = 8  
b = 9  
  
c = a - b  
print(c)
```

-1

```
In [80]: ▶ a = 8  
b = 9  
  
c = a * b  
print(c)
```

72

```
In [85]: ▶ a = 19  
b = 9  
  
c = a / b  
print(c)
```

2.1111111111111111

```
In [84]: ▶ a = 19  
b = 9  
  
c = a // b  
print(c)
```

2

```
In [86]: a = 19
b = 9

c = a % b
print(c)
```

1

```
In [87]: a = 2
b = 6

c = a ** b
print(c)
```

64

```
In [ ]:
```

```
In [ ]: lst = [1,2,3,4,5,6,]
```

```
In [92]: print(list(range(1,101)))
```

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100]

```
In [ ]:
```

Comparison Operators

```
In [2]: a = 5
b = 6

if a == b:
    print('True')
else:
    print('False')
```

False

```
In [1]: 5 == 6
```

Out[1]: False

```
In [4]: a = 5
b = 5

if a == b:
    print('True')
else:
    print('False')
```

True

```
In [3]: a = 5
b = 6

if a != b:
    print('True')
else:
    print('False')
```

True

In []: ▶

In [6]: ▶

```
a = 6
b = 6

if a < b:
    print('True')
else:
    print('False')
```

False

In [9]: ▶

```
a = 6
b = 6

if a <= b:
    print('True')
else:
    print('False')
```

True

In []: ▶

In [10]: ▶

```
a = 6
b = 6

if a > b:
    print('True')
else:
    print('False')
```

False

In []: ▶

In [12]: ▶

```
a = 6
b = 4

if a >= b:
    print('True')
else:
    print('False')
```

True

In []: ▶

Logical Operators

```
In [22]: ▶ a = 6
          b = 0
          c = -4

          if a > 0 and b < 0:
              print('True Statement')
          else:
              print('False Statement')

          # print('\t')
          print('\n')

          if a > 0 and c < 0:
              print('True Statement')
          else:
              print('False Statement')
```

False Statement

True Statement

```
In [ ]: ▶
```

```
In [23]: ▶ a = 6
          b = 0
          c = -4

          if a > 0 or b < 0:
              print('True Statement')
          else:
              print('False Statement')

          print()

          if a > 0 or c < 0:
              print('True Statement')
          else:
              print('False Statement')
```

True Statement

True Statement

```
In [ ]: ▶
```

```
In [27]: ▶ a = 9

          if not(a%3==0):
              print('True')
          else:
              print('False')
```

False

```
In [ ]: ▶
```


Assignment Operators

```
In [28]:  x = 6  
          x
```

Out[28]: 6

```
In [29]:  x = 6  
          x += 1    # x = x + 1 = 6 + 1 = 7  
          x
```

Out[29]: 7

```
In [32]:  x = 6  
          # x = x - 1  
          x -= 1  
          x
```

Out[32]: 5

```
In [34]:  x = 6  
          x *= 2  
          x
```

Out[34]: 12

```
In [35]:  x = 6  
          x /= 2  
          x
```

Out[35]: 3.0

```
In [36]:  x = 6  
          x //= 2  
          x
```

Out[36]: 3

```
In [37]:  x = 15  
          x /= 2  
          x
```

Out[37]: 7.5

```
In [38]:  x = 15  
          x //= 2  
          x
```

Out[38]: 7

```
In [39]:  x = 15  
          x %= 2  
          x
```

Out[39]: 1

```
In [41]:  x = 5  
          x **= 3  
          x
```

Out[41]: 125

```
In [ ]:  
```

Membership Operators

```
In [45]: ▶ lst1 = [1,2,3,4,5,6]
          lst2 = [2,4,6,8,10,12]

          for item in lst1:
              #     print(item)
              if item in lst2:
                  print(item)

          print()

          for item in lst1:
              #     print(item)
              if item not in lst2:
                  print(item)
```

```
2
4
6

1
3
5
```

```
In [ ]: ▶
```

Identity Operators

```
In [46]: a = 5  
b = 5  
  
print(a is b)
```

True

```
In [47]: a = 5  
b = 6  
  
print(a is b)
```

False

```
In [ ]:
```

```
In [48]: a = 5  
b = 5  
  
print(a is not b)
```

False

```
In [49]: a = 5  
b = 6  
  
print(a is not b)
```

True

In []: ▶

In []: ▶

In []: ▶

In []: ▶

In []: ▶

In []: ▶

In []: ▶

In []: ▶

In []: ▶